

Analyzing and Mitigating Motor Vehicle Crashes in New York City

Data-Driven Insights and Recommendations for Safer Roads in New York City

Motor vehicle crashes are a significant concern in New York City, particularly for vulnerable road users like pedestrians and cyclists. This study focuses on understanding key factors contributing to crashes, identifying high-risk areas and times, and providing data-driven recommendations to improve road safety. The research was guided by four key questions: factors influencing fatal crashes across boroughs, the most dangerous times and locations for pedestrian-involved crashes, the predictability of crash severity based on various factors, and intersections posing high risks for cyclists.

The analysis utilized the NYC Motor Vehicle Collisions dataset, applying exploratory data analysis, geospatial mapping, and predictive modeling techniques. Key findings revealed that fatal crashes were concentrated in Manhattan and Brooklyn, with late-night hours posing a higher risk. Pedestrian-involved crashes peaked during rush hours, especially in busy commercial areas. High-risk intersections for cyclists were identified, with a lack of protected bike lanes contributing to increased crash severity. Additionally, a Random Forest model achieved 85% accuracy in predicting crash severity, highlighting vehicle type and time of day as the most significant predictors.

Based on these insights, several recommendations were proposed to improve road safety. These include targeted traffic enforcement during high-risk periods, infrastructure improvements such as protected bike lanes and raised crosswalks, and public awareness campaigns focusing on pedestrian and cyclist safety. Enhancing data collection on road conditions and minor crashes was also suggested to support future research and policy-making efforts.

Outstanding Questions

1. How do specific weather conditions influence crash severity?
2. How does real-time traffic density impact crash frequency?